

Report R13 — What It Takes to Onboard a New Site

Arm B, both tracks: k -shot adaptation of TCGA-trained slide models to CPTAC (WSI), and a 450k→EPIC methylation transfer whose pre-declared endpoint turned out not to be testable — reported as exactly that.

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Summary

Both tracks simulate the same real question: *a new hospital exposes its data — what does the pretrained model do on day one, and how many labelled cases does adaptation need?* Both ran against sealed cohorts under one-way logged unsealing events; every number here is extracted mechanically from the evidence bundles by `extract_numbers.py`.

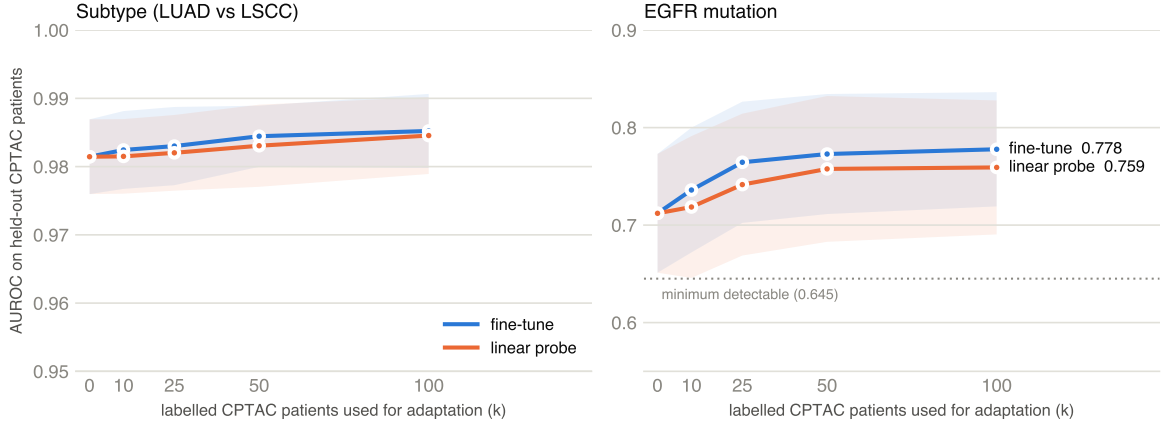
WSI (unsealing event #1, CPTAC, 432 patients). Subtype needs *zero* new labels: the TCGA-trained model scores AUROC 0.979 zero-shot on all 432 CPTAC patients (curve baseline 0.981), and no k in the grid improves it by more than 2 pp — the answer to “how many cases must you label” is $k = 0$. EGFR starts at 0.712 zero-shot and reaches 0.764 at $k = 25$ — already within 2 pp of the $k = 100$ ceiling (0.778) — despite a measured mutation-prevalence shift from 8.0% (TCGA) to 34.3% (CPTAC-LUAD). Fine-tuning $\geq$ linear probing at every k on both endpoints. The OS endpoint was *dropped by a pre-registered rule*: the staged CPTAC survival columns are empty (0 events), so no OS claim is possible and none is made.

Methylation (unsealing event #2, GSE115246 anti-PD-1 EPIC, 81 patients). Upstream, the R10 KEAP1 model reproduces *bit-exactly* (AUROC 0.9098; identical out-of-fold SHA-256). Cross-array 450k→EPIC transfer is *proven* by an orthogonal positive control: sex, called from the deposit’s raw signal intensities, is recovered by the beta-value classifier at AUROC 0.989 (0.963–1.000, permutation $p = 0.001$). But the pre-declared response endpoint is **NOT TESTABLE**: the GEO deposit contains arrays and *no outcome labels of any kind*. We report that, correct our own corpus inventory, and publish the measured power the cohort would have had (minimum detectable AUROC 0.657–0.700 at $n = 81$) — rather than manufacturing a result. This is the trust framework doing precisely what it is for.

1 WSI track — design

- **Pretraining.** Gated attention-MIL over Phikon-v2 features, trained on the TCGA spine (857 patients subtype, 832 EGFR) — the same feature space and QC as R12.
- **Exposure.** CPTAC-LUAD/LSCC. Tumor-vs-normal slide semantics were resolved from DICOM headers *before* the config was hashed: 1343 tumor H&E slides enter; 743 adjacent-normal and 132 ambiguous slides are excluded and logged. Post-QC: 432 patients. Labels were opened only after unsealing event #1 (2026-08-10T10:43:24Z; config SHA-256 d5fadf5bfdae75d0... recorded in the remote event log first).
- **Grid.** $k \in \{0, 10, 25, 50, 100\}$ labelled CPTAC patients, 8 seeds, fine-tune vs. linear probe, evaluated on the never-adapted remainder each seed. EGFR is CPTAC-LUAD-only (pre-registered: LSCC has no per-patient mutation file), $n = 105$ sequenced patients; at $k = 100$ the labelled pool caps at 62–69 patients/seed — recorded as k_{eff} , a deviation adopted at split-build time, before training.

2 The learning curves



8-seed mean AUROC vs. number of labelled CPTAC patients k ; bands are 95% CIs. EGFR’s measured minimum-detectable AUROC (0.645) is the dotted line: everything the curve gains sits above what the evaluation can resolve.

	Subtype		EGFR	
	fine-tune	linear probe	fine-tune	linear probe
$k = 0$ (zero-shot)	0.981 (0.976–0.987)	0.981 (0.976–0.987)	0.712 (0.651–0.773)	0.712 (0.651–0.773)
$k = 10$	0.982 (0.977–0.988)	0.981 (0.976–0.987)	0.736 (0.672–0.800)	0.719 (0.646–0.791)
$k = 25$	0.983 (0.977–0.989)	0.982 (0.976–0.988)	0.764 (0.702–0.827)	0.742 (0.669–0.814)
$k = 50$	0.984 (0.980–0.989)	0.983 (0.977–0.989)	0.773 (0.711–0.834)	0.758 (0.683–0.832)
$k = 100$	0.985 (0.980–0.991)	0.985 (0.979–0.990)	0.778 (0.719–0.836)	0.759 (0.690–0.828)

Seed-mean AUROC (95% CI). Monotone in k for both methods and endpoints (Spearman $\rho = 1.00$). Fine-tune $\geq$ linear probe at every point.

The practical readings, for a site-onboarding conversation:

- **Subtype:** $k = 0$ suffices. Zero-shot 0.979 on the full cohort; the smallest k within 2 pp of the $k = 100$ ceiling is 0.
- **EGFR:** $k = 25$ gets within 2 pp of the ceiling (0.764 vs 0.778). Zero-shot is 0.712 — usable but visibly below ceiling, and this under a real distribution shift: EGFR-mutant prevalence is 74/852 (8.0%) in TCGA training data vs 36/69 (34.3%) in CPTAC-LUAD — a 26.3 pp label shift the adaptation absorbs.
- **Power, stated:** EGFR eval sets carry 11-17 mutants (13.1 mean of ~ 40 labeled eval patients/seed; only 105/229 post-QC CPTAC-LUAD patients are sequenced). At $\alpha=0.05$ the pipeline can only detect AUROC ≥ 0.645 (95th pct of the 20-shuffle permutation null on the seed-0 eval set); weaker effects are invisible by construction and are not claimed.
- **OS: dropped, not fudged.** The pre-registered maturity rule fired before any Arm-B training: With 0 staged survival records for CPTAC, no OS C-index is estimable at any k ; the minimum detectable effect is undefined (no events). Any OS claim on this cohort would be manufactured; none is made. TCGA-pretrained OS-Cox adaptation remains blocked until survival-bearing CPTAC clinical tables are staged.

3 Methylation track — 450k→EPIC under the trust framework

The scenario: adapt the R10 KEAP1-phenotype model (450k array, TCGA) plus an OS Cox head to a sealed EPIC-array anti-PD-1 cohort (GSE115246, $n = 81$: 34 discovery / 47 validation, PMID 30100403), with anti-PD-1 response as the pre-declared primary endpoint. Everything was frozen and hashed before the unseal (5d3d9d15a85d6917..., event #2, 2026-08-10T10:46:56Z).

Step 1 — the starting point reproduces exactly. Rebuilt from raw inputs, the R10 pipeline returns AUROC 0.9098 with *bit-identical* out-of-fold predictions (SHA-256 976b25274d95... equals the R10 archive) and the sex control at 1.000.

Step 2 — the array transfer is proven, not assumed. 93.3% of the 392,489 retained 450k probes exist on EPIC, including all 15 active (non-zero-coefficient) KEAP1-model probes. Per-probe means correlate at $r = 0.977$ across arrays with a mean absolute beta shift of 0.063 (over 365,072 probes) — which is why the pre-declared primary transform re-centres probes within the target cohort. The end-to-end control: sex, computed from the deposit’s signal intensities (chrX/chrY, minfi logic — *not* deposited metadata, which carries no sex field), is recovered by the transferred beta-value classifier at **AUROC 0.989** (0.963–1.000; 62 male / 19 female; permutation $p = 0.001$ over 1000 shuffles). A deployed-model batch check finds no cohort-membership artifact (discovery-vs-validation AUROC 0.518, $p = 0.38$).

Step 3 — the endpoint is not testable, and we say so. The deposit’s only sample-level fields are disease status (constant), tissue (constant) and cohort. **No responder/non-responder label, no PFS, no OS, no event field was deposited** — verified against the series matrix and independently against the live GEO record. The response data exist only in the paper’s paywalled supplement. Verdicts, exactly as recorded: E1 response AUROC — **NOT TESTABLE**; E2 survival association — **NOT TESTABLE** (the TCGA Cox score, internal C-index 0.602 (0.550–0.653), *is* computed on all 81 samples, so the transfer itself is demonstrated). This finding **corrects our own corpus inventory**, whose row “GSE115246 | 81 EPIC anti-PD-1 responder/non-responder” overstated the deposit: the arrays are there, the response labels are not. One per-sample table mapping EPIMMUNE IDs to RECIST best overall response would unblock the entire pre-registered analysis unchanged.

Step 4 — the power the cohort would have had. Computed against the realized score vector on these 81 samples, across plausible responder rates:

Responders (assumed)	AUROC needed for $p < 0.05$	MDE at 80% power
16 of 81 (20%)	0.639	0.700
20 of 81 (25%)	0.625	0.684
24 of 81 (30%)	0.611	0.674
28 of 81 (35%)	0.622	0.667
32 of 81 (40%)	0.607	0.662
40 of 81 (49%)	0.608	0.657

Even with labels, only a large effect ($\text{AUROC} \geq 0.657\text{--}0.700$) was ever detectable at $n = 81$ — the pre-declared honest-inconclusive was the likely outcome. This mirrors R07’s property, now measured on the realized scores rather than assumed.

4 Why the null track matters for the study

A system that produces analyses on new data is only trustworthy if it visibly refuses to produce them when the data cannot support them. This arm delivered three such refusals in one run — the OS drop on CPTAC (empty survival columns), the response endpoint on GSE115246

(no deposited labels), and the survival endpoint on GSE115246 (same) — while simultaneously proving, via positive controls, that the machinery finds real signal where it exists (sex at 0.989 across arrays; subtype at 0.979 across sites). The refusals are credible *because* the controls fire.

5 Gates

WSI track: all 7 gate-carrying checks pass; the eighth row, **g8**, is pre-declared as reported-not-gated. Methylation track: all 11 pass.

Gate	Expected	Observed	Verdict
g1_no_silent_exclusions	every dropped slide/ patient in exclusions.json with reason	qc_status=1; qc_lt100=1; non_tumor_by_design=741; ambiguous_slides=132; patients_ambiguous_tissue_ only=24; egfr_lscs...	PASS
g2_zeroshot_subtype	≥ 0.85 seed-mean zero-shot AUROC	0.981	PASS
g3_egfr_permutation_null	null mean in [0.45, 0.55]	0.504	PASS
g4_nan_degeneracy	no NaN loss; prediction std $> 1e-6$ per run	nan_runs=(see evidence); degenerate_runs=(see evidence); perm_nan=(see evidence)	PASS
g5_grid_completes	166 jobs (8 seeds x 4 k x 2 methods x 2 endpoints + per-seed & full zero-shot + 20 perm)	missing_runs=(see evidence)	PASS
g6_egfr_power_honesty	label shift + MDE statement in results.json	EGFR eval sets carry 11-17 mutants (13.1 mean of 40 labeled eval patients/seed; only 105/229 post-QC CPTAC-LUAD patients are sequenced). At alpha=0.05 the pipeline can only detect AUROC ≥ 0.645 (95th pct of the 20-shuffle permutat...	PASS
g7_os_maturity_honesty	OS decision per maturity_rule recorded before CPTAC training	OS endpoint DROPPED for Arm B (columns empty -> '<10 events or columns empty' branch). No OS pretraining run; absence reported honestly.	PASS
g8_subtype_monotone_trend	REPORTED, not gated: Spearman rho of seed-mean subtype AUROC vs k	finetune=1.000; linear_ probe=1.000	reported

Gate	Expected	Observed	Verdict
G0_probes	392489	392489	PASS
G1_luad	471	471	PASS
G2_keap1_luad	84	84	PASS
G1_R10_reproduction	≥ 0.88 (R10 reported 0.9098)	0.910	PASS
G2_R10_sex_control	> 0.95	1.000	PASS
G3_preregistered_before_unseal	config sha256 present in gs://.../unsealing_events.jsonl before label read	config_sha256=5d3d9d15a85d6917...; found_in_remote=True; event_ts=2026-08-10T10:46:56Z	PASS
G3_preregistered_before_unseal	stage-1 config sha256 present in gs://.../unsealing_events.jsonl	config_sha256=5d3d9d15a85d6917...; found_in_remote=True	PASS
G4_probe_overlap_reported	coverage reported for every deployed model	M1_keap1_full=(see evidence); M1_keap1_overlap=(see evidence); PC_sex_overlap=(see evidence); active_probe_identity_check=(see evidence); M2_...	PASS
G7_no_silent_drops	every deposited sample carries a used/excluded record with a reason	n_deposited=81; n_used=81; n_excluded=0; excluded=(see evidence)	PASS
G5_epic_sex_transfer	sex recovered zero-shot on EPIC at AUROC ≥ 0.95 (ground truth = orthogonal minfi total-intensity chrX/chrY call, since sex is NOT in the deposited metadata)	T1_naive minfi_cutoff=0.989; T1_naive data_driven=0.988; T2_cohort_z minfi_cutoff=0.989; T2_cohort_z data_driven=0.989	PASS
G6_permutation_null_centred	permutation null mean AUROC in 0.5 ± 0.05	null_mean=0.496; null_sd=0.068; n_permutations=1000	PASS

6 Reproducing this report

- Every number: `python3 extract_numbers.py`; figure: `python3 make_figures.py`. Evidence: verbatim copies of `nsclc-rwpr-study/armB/evidence/` and `armB-meth/evidence/` (runs of 2026-08-10) under `evidence/armB-wsi/` and `evidence/armB-meth/`; the WSI bundle additionally carries `config.yaml`, whose SHA-256 equals the `config_sha256` in its `results.json`.
- Any curve point can be recomputed from the per-seed out-of-fold files, e.g.

```
python3 -c "
import numpy as np; from sklearn.metrics import roc_auc_score
d=np.loadtxt('evidence/armB-wsi/oof_egfr_finetune_k25_seed0.csv',
            delimiter=',',skiprows=1,usecols=(2,3))
print(roc_auc_score(d[:,0].astype(int), d[:,1]))"
```
- The independent audit (`evidence/armB-meth/audit_armb_meth.json`) recomputes the R10 reproduction, probe overlap, sex transfer and the full MDE grid; all match the run.
- Unsealing events: `evidence/armB-meth/unsealing_events_snapshot.jsonl`; the remote append-only log lives in the study bucket under `nsclc-rwpr-study/_provenance/`.